Amazon "SageMaker AI"

- ✓ Ent – IDE → fully infrst
- ✓ Fully-Managed ML & AI Platform
- ✓ Build, Train, fine-tune, deploy ML/AI Models at Scale

ML Lifecycle	(Sagemaker AI – ML factory)
(1) Data Preparation	Data Wrangler
(2) Model Building	Notebooks, Autopilot, Jumpstart
(3) Training	Managed Tech Cluster
(4) Tuning	Hyperparameter tuning
(5) Deployment	Real-time endpoint
(6) Monitoring	Model Monitor
(7) Governance	Model Cards CARDS

Core Capabilities – SageMaker AI

(1) Data Preparation → Prepare ML Datasets.
- ✓ Data Wrangler → Data cleaning & feature engineering
- ✓ Ground TRUTH → Data labeling
- ✓ Feature Store → Reusable ML features.

(2) Model Development
- * ✓ SageMaker Studio (ML IDE)
- * ✓ Jupyter Notebooks
- ✓ R Studio
- ✓ VS Code-based editor

(3) Model TRG
- ✓ Built-in Algs
- ✓ Custom Models
- ✓ Deep Learning frameworks (PyTorch, TensorFlow)
- ✓ Distributed Trg Clusters

(4) AutoML & prebuilt Models
- ✓ Autopilot → Automatic Model Creation
- ✓ Jumpstart (GenAI) → Pretrained Model & templates (FM/LLMs)
- ✓ Canvas → No-Code ML

(5) Model Deployment
- ✓ Real-time Inference endpoints
- ✓ Batch-Inference
- ✓ Serverless-Inference
- ✓ Edge Deployment

* (6) MLOPS (ML operations)
  - ↳ Enterprise ML Lifecycle
- ✓ Model pipeline
- ✓ Model Registry
- ✓ Experiment Tracking
- ✓ Debugger
- ✓ Model Monitor

(7) Generative AI Support
- ✓ Custom LLM Tech
- ✓ Adv ML experiments
- ✓ fine-tune FMs
- ✓ RAG Architecture

SageMaker Components

- * (1) SageMaker Studio → ML development IDE
- (2) Data Wrangler → Data Prep
- (3) Feature Store → Feature Management
- * (4) Pipelines → ML CI/CD
- (5) JumpStart → Pretrained Models
- (6) Autopilot → Auto ML
- * (7) Model Monitor → Production
- (8) Clarify → Bias detection
- (9) Grand Truth → Data labeling

Yes = +1, No = +2

SageMaker AI → Domain → Top-level ORG Container
↳ SageMaker Resources

AWS A/c → SageMaker AI → Domain → users → Projects → Apps.

Domain
- users
- Auth setting
- Studio env
- Storage Config
- Security policies

| Domain — ORG | Team

User profile → Studio

User Profile → SageMaker User
↳ Apps (Actual Running env)

SageMaker Domain
- User Profile → (S3)
  - Apps
  - Private files
  - Notebooks
- Spaces
  - ↳ Shared apps (Team)
  - ↳ Shared Storage
- Projects
  - ↳ Pipelines
  - ↳ Model Registry
- → Storage (EFS) ✓

Lifecycle Config
↳ Startup scripts

Domain entity Relationships

User Profile	→ Domain
Space	→ Domain
✓ App	→ User / Space
Project	→ Domain
Storage	→ Domain

Real-time Enterprise Design

Domain: Ent-ML-Domain

Users: Data Sci
ML Eng
MLOPS Team

Spaces: NLP Team Space
vision Team space

Project: Recommendation Engine
Fraud Detection

Apps: Jupyter Lab
VS Code
RStudio

Data → Where? S3 + Datalake

How? Data Preparation ⇒ SageMaker AI Data Wrangler

Feature Groups ⇒ SageMaker Feature Store